DON'T PANIC

Automated Plant Phenotyping for Root System Architecture

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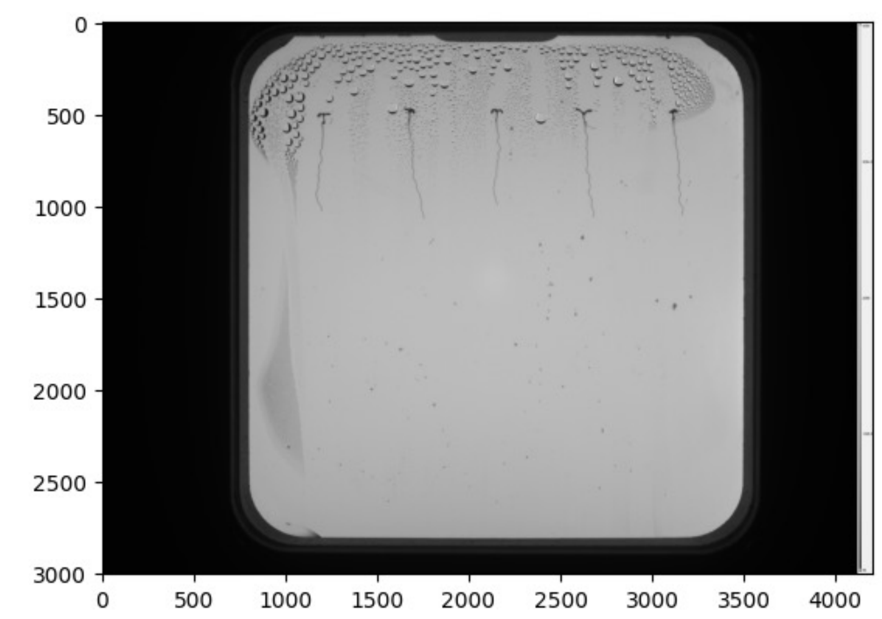
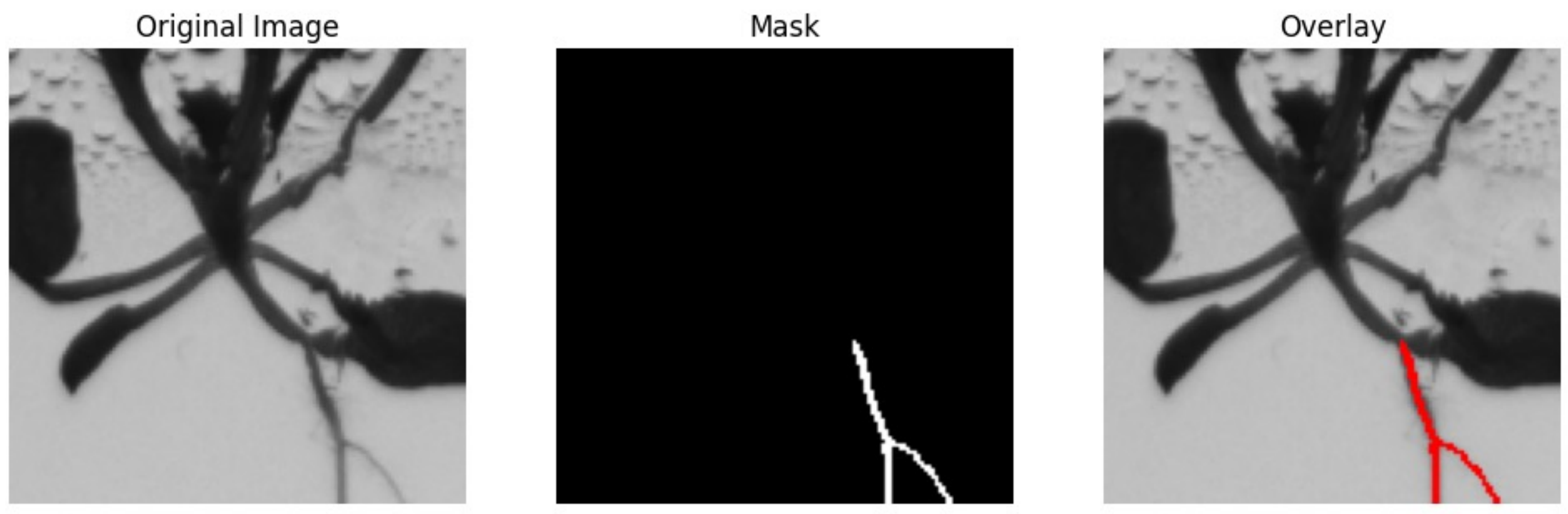
Raw Petri Dish Image → U-Net Segmentation → Skeleton Extraction → Dijkstra Measurement → Root Tip Mapping → Robot Simulation

1. Raw petri dish image



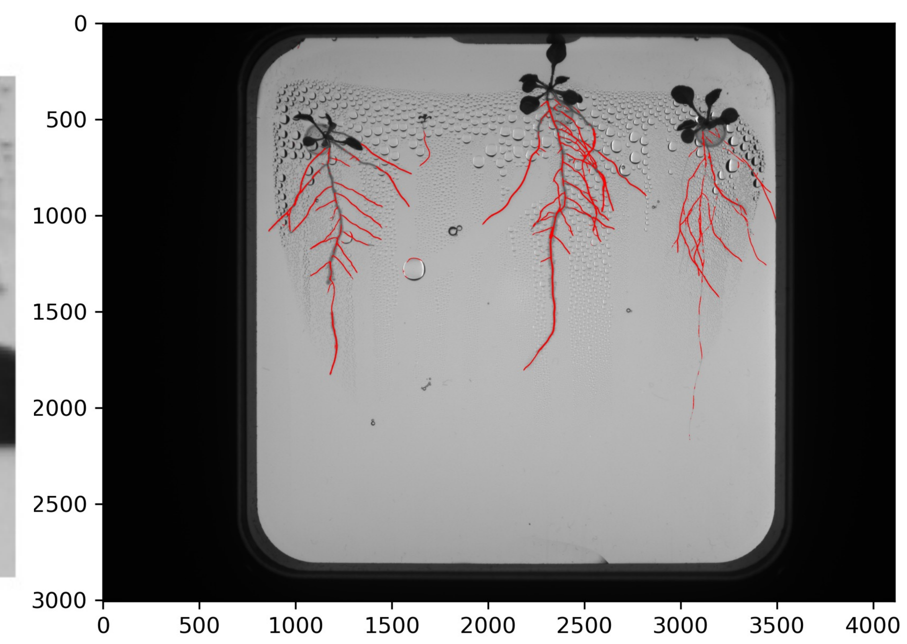
Input: 4000×3000 px image from NPEC Hades imaging system

2. U-Net segmentation patches



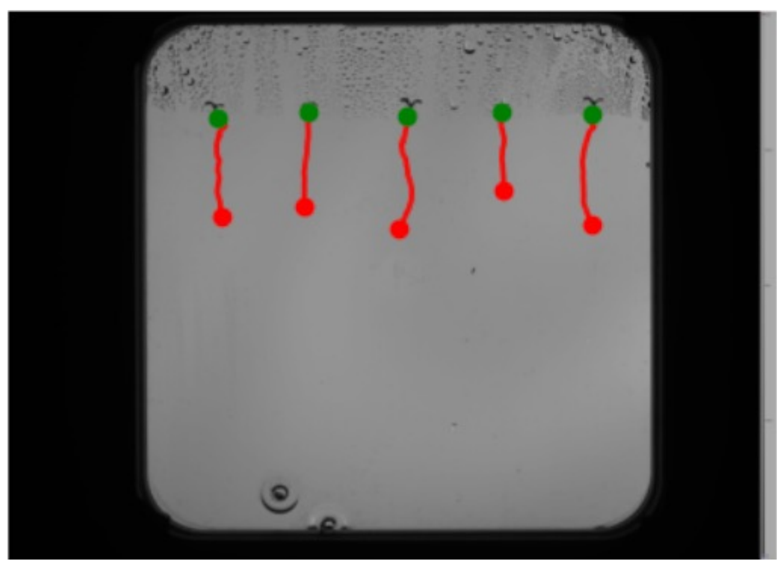
256×256 patches processed by U-Net, reassembled into full mask

3. Outputted Skeleton



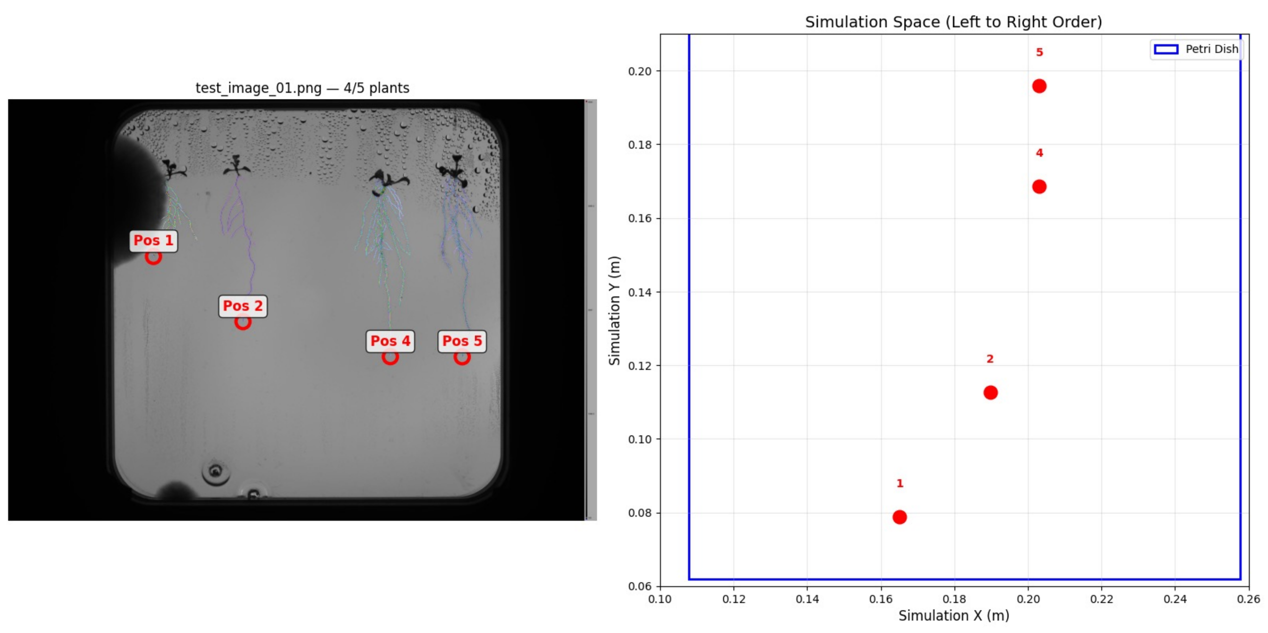
Inference using U-Net with 0.83 F1 Score

4. Dijkstra Measurement



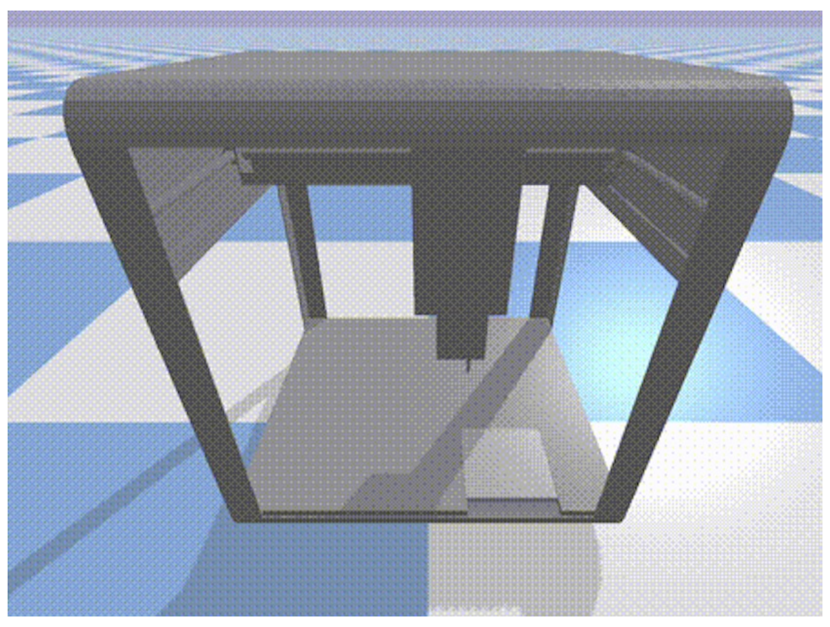
19 Image Kaggle dataset sMAPE 3.2%

5. Root tip mapping



Dijkstra endpoint → pixel-to-meter → robot workspace coordinates

6. Robot simulation

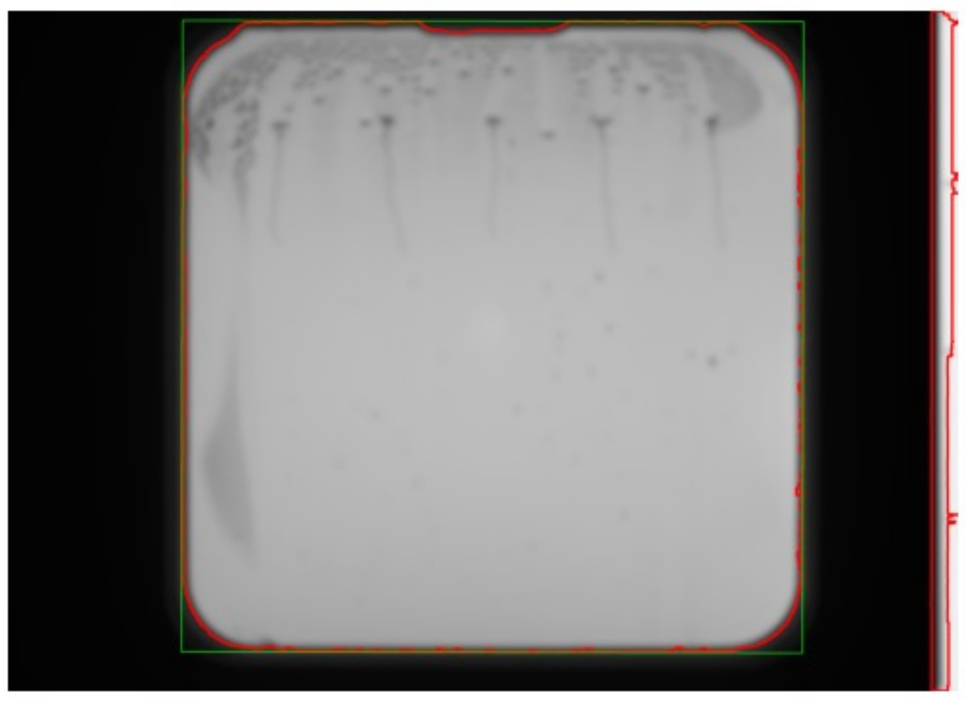


PID-controlled pipette placement with sub-mm accuracy

42 *Don't Panic — the answer to root measurement was simpler than we thought*

1 Shape-based noise removal

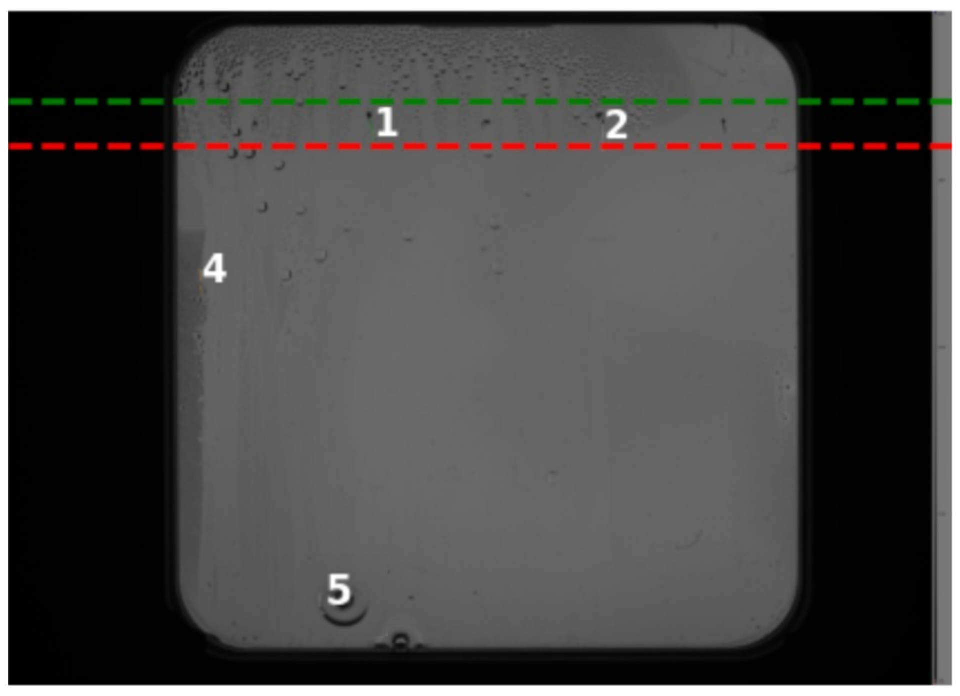
Problem: Water bubble residuals left arc-shaped artifacts in segmentation masks



Fix: Filter components by shape — arcs are elongated, roots are not

2 XY band filtering

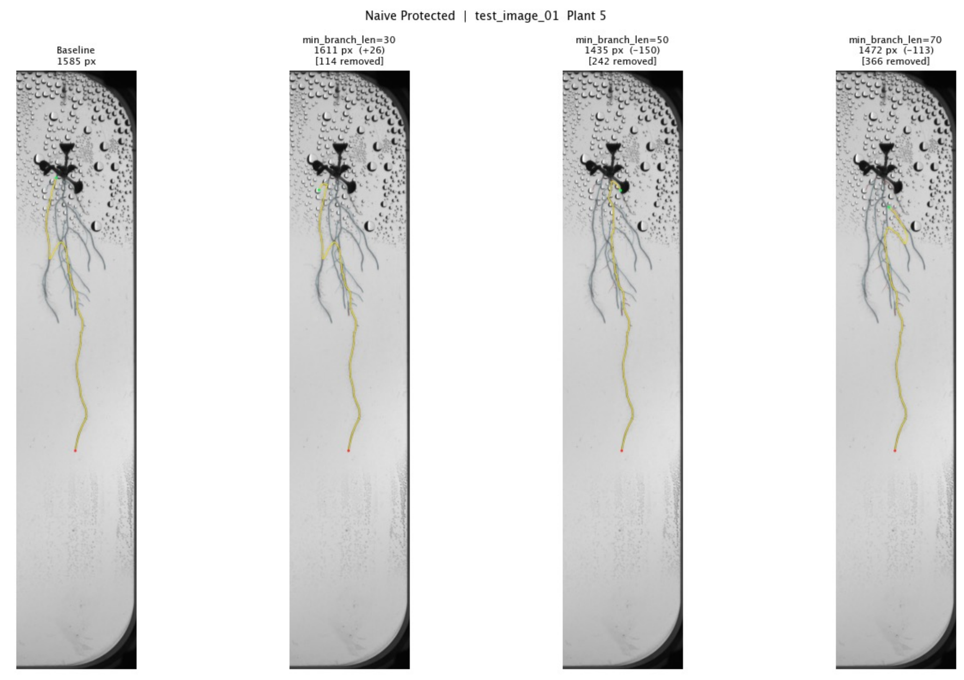
Problem: Segmentation included noise outside the plant's vertical growth zone



Fix: Define per-plant XY bands from seed positions. Discard "plants" outside the band

3 Heuristic pruning

Problem: Dijkstra followed lateral roots or picked wrong start after branch removal



Fix: Remove smaller roots, make longest path follow natural paths